D06 · Freight Management & Mode Selection

03 · Freight consolidation, LTL vs. FTL & load optimization

Supply Chain Reference Architecture v2.2 · mrsupplychain.ai

1 WHAT IT IS

Consolidation combines small shipments into fuller, less frequent loads; the LTL vs. FTL decision compares paying per shipment against paying for the whole truck; load optimization fills each trailer by weight and cube without breaking stacking or delivery-sequence rules.

2 WHY IT MATTERS

Transport has strong economies of scale: **the cost per unit falls sharply as the load fills**. Consolidation captures those economies, but it trades against responsiveness — holding orders to fill a truck delays some customers **(Bowersox, Closs and Cooper — Supply Chain Logistics Management, textbook)**. The design question is how long you can hold, and for whom.

3 HOW IT IS DONE

1. **Map shipment sizes by lane.** Find the lanes where many small LTL shipments go to the same region on the same days.
2. **Define consolidation windows by customer segment.** Strategic customers keep daily service; the rest consolidate into scheduled departures.
3. **Find the LTL–FTL breakeven per lane.** Above a weight or pallet threshold, a full or multi-stop truckload beats summed LTL charges.
4. **Optimize the load, then measure fill rate.** Use load-building tools for weight and cube, and track trailer utilization as a standing KPI.

IN PRACTICE

Build the destination and volume map before designing the consolidation strategy

Without accurate volume and destination data the consolidation design is based on assumptions that usually prove wrong.

Consolidation requires order cut-off discipline

Moving from daily LTL to 2–3 FTL per week forces the commercial team to accept earlier order cut-offs. Without that discipline, consolidation creates late shipments.

Use milk-runs for low-volume destinations

A well-designed milk-run is frequently more efficient than either pure LTL or forced FTL for low-volume points.

ILLUSTRATIVE CASE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case — FMCG company, 840 LTL shipments/month — 68% of destinations were candidates for consolidation into FTL.

Destination	Before	After
North DC (3.2 t/day)	12–15 LTL/week	3 FTL/week (-28% cost)
South DC (2.8 t/day)	10–12 LTL/week	2–3 FTL/week (-22% cost)
Regional stores (<0.5 t/week)	LTL	Milk-run (no change)

Result: 840 LTL -> 320 shipments (140 FTL + 180 LTL). Freight cost -21%. Management complexity -62%.

HOW IT IS MEASURED

Truck Fill Rate %

(Actual volume or weight / Truck capacity) × 100

Benchmark: >85% for FTL · >70% for large LTL

LTL to FTL Conversion Rate %

(Volume converted from LTL to FTL / Prior LTL volume) × 100

Benchmark: Each point of conversion typically yields 15–30% savings on that volume

WHAT YOU WOULD USE

Load Optimization

SAP TM — Load Building — 3D load optimization and FTL/LTL decision engine.

Blue Yonder / Ortec — Advanced 3D packing algorithms and multi-order consolidation.

4 SO WHAT

Fill the truck on purpose. Find the LTL–FTL breakeven by lane, set consolidation windows by customer segment, and make trailer utilization a KPI someone owns.

From the SCRA v2.2 reference architecture · mrsupplychain.ai · © 2026 Ismael López · Mr. Supply Chain® México